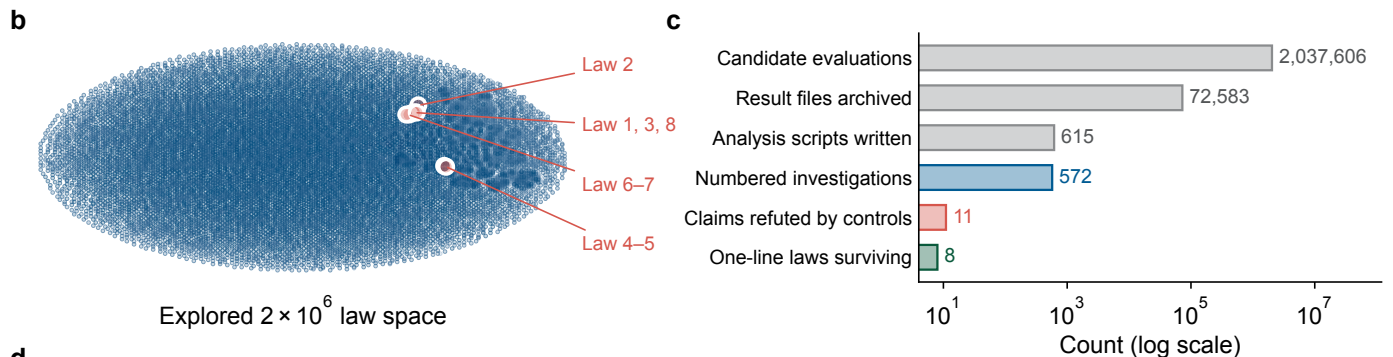
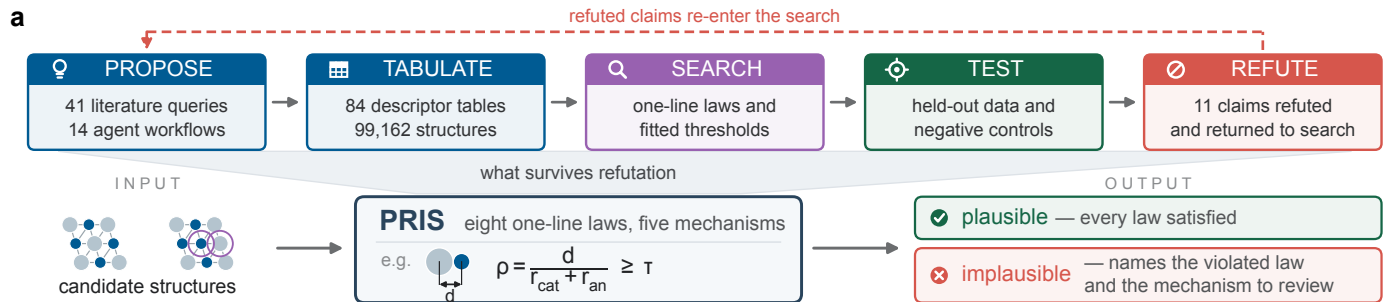




d

	Set 1 hard-sphere floor	Set 1' two-sided window	Set 2 rigid-ion lattice	Set 3 ionic network	Set 4 crystal chemistry
Law 1 $\rho \geq \tau$ shortest cation-anion contact, relative to the radius sum	0.735	0.735	0.804	0.804	0.804
Law 2 if $f_i > 0.50$ then $\rho \leq 1.05$ ionicity condition: an upper bound applied only where the ionic model holds	—	●	—	—	—
Law 3 if mean anion CN ≤ 3.333 then mean $d/(r_{\text{cat}} + r_{\text{an}}) \leq 1.081$ low-coordination structures may not also be loosely packed on average	—	—	●	●	●
Law 4 range of $E_M/ z $ across sites ≤ 31.45 no site may be electrostatically far out of line with the others	—	—	●	●	●
Law 5 $\max_i E_M(i) \leq 15.17$ no single site may have an implausible Madelung energy	—	—	●	●	●
Law 6 if $f_i > 0.55$ then no like-charge bonds charge topology: ionic compounds do not bond like to like	—	—	—	●	●
Law 7 inequivalent sites / sites $\leq 2/3$ distinct-site bound consistent with a preference for simpler structures	—	—	—	—	●
Law 8 mean relative bond-valence deviation ≤ 0.7143 a permissive tail bound on the valence sum, the quantity of rule 2	—	—	—	—	●
satisfaction (experimental structures)	0.9919	0.9894	0.9579	0.9171	0.8180
damage detection	0.2890	0.3837	0.6121	0.7004	0.9111

